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53. Sales Transaction Analysis

Files Required

- Sales.csv
- Stores.csv
- Sales.json
- Sales.xml

```
import pandas as pd

# Read CSV files
sales = pd.read_csv("sales.csv")
stores = pd.read_csv("stores.csv")

# Read JSON file
sales_json = pd.read_json("sales.json")

# Read XML file
sales_xml = pd.read_xml("sales.xml", parser="etree")

# Display input data
print("Sales Data:")
print(sales.head())

print("\nStores Data:")
print(stores.head())

print("\nSales Information:")
print(sales.info())

# Handle missing values
sales.fillna(0, inplace=True)

# Remove duplicate records
sales.drop_duplicates(inplace=True)

# Convert data type
sales["SalesAmount"] = sales["SalesAmount"].astype(float)
```

```
# Remove extra spaces from column names
sales.columns = sales.columns.str.strip()
stores.columns = stores.columns.str.strip()

# Merge Sales and Stores using StoreID
merged = pd.merge(sales, stores, on="StoreID")

# Display merged data
print("\nMerged Data:")
print(merged)

# Summary statistics
print("\nSummary Statistics:")
print(merged.describe())

# Sales by category
summary = merged.groupby("Category")["SalesAmount"].sum()

print("\nSales by Category:")
print(summary)

# Highest sales category
print("\nHighest Sales Category:")
print(summary.idxmax())

# Export cleaned dataset
merged.to_csv("Cleaned_Sales.csv", index=False)

print("\nCleaned dataset exported successfully.")
```

Sample Data

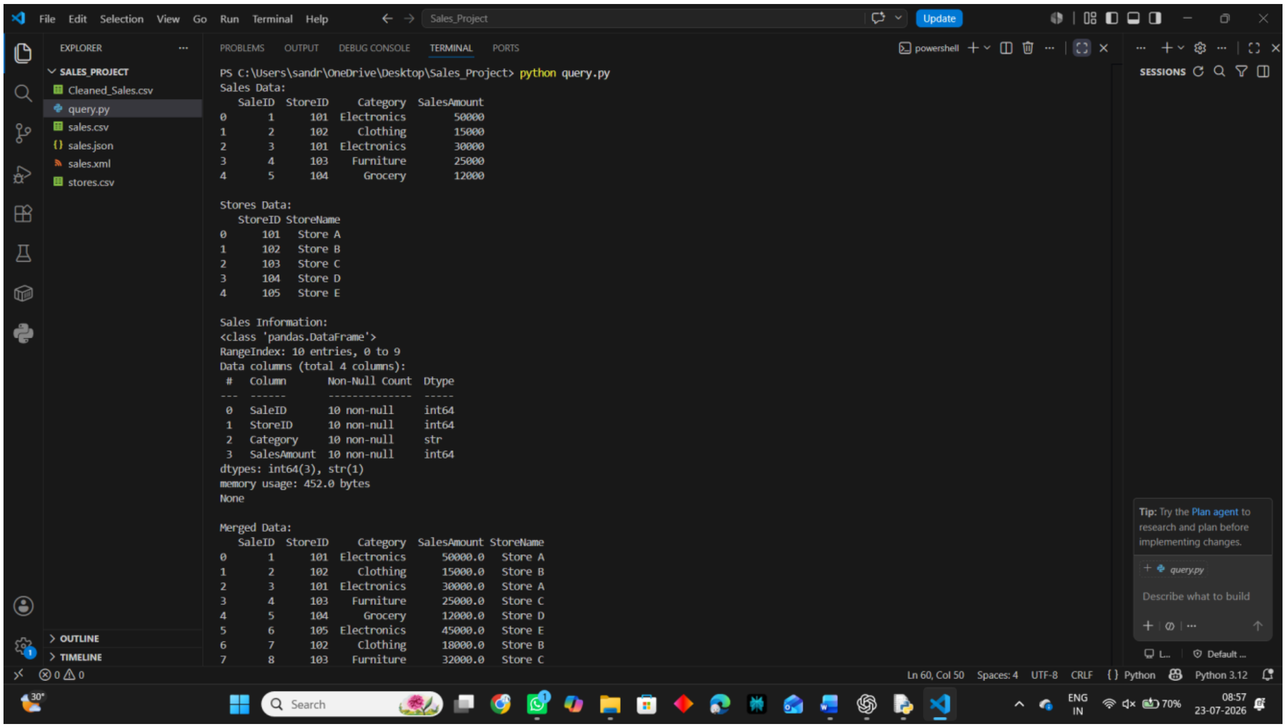
Sales.csv

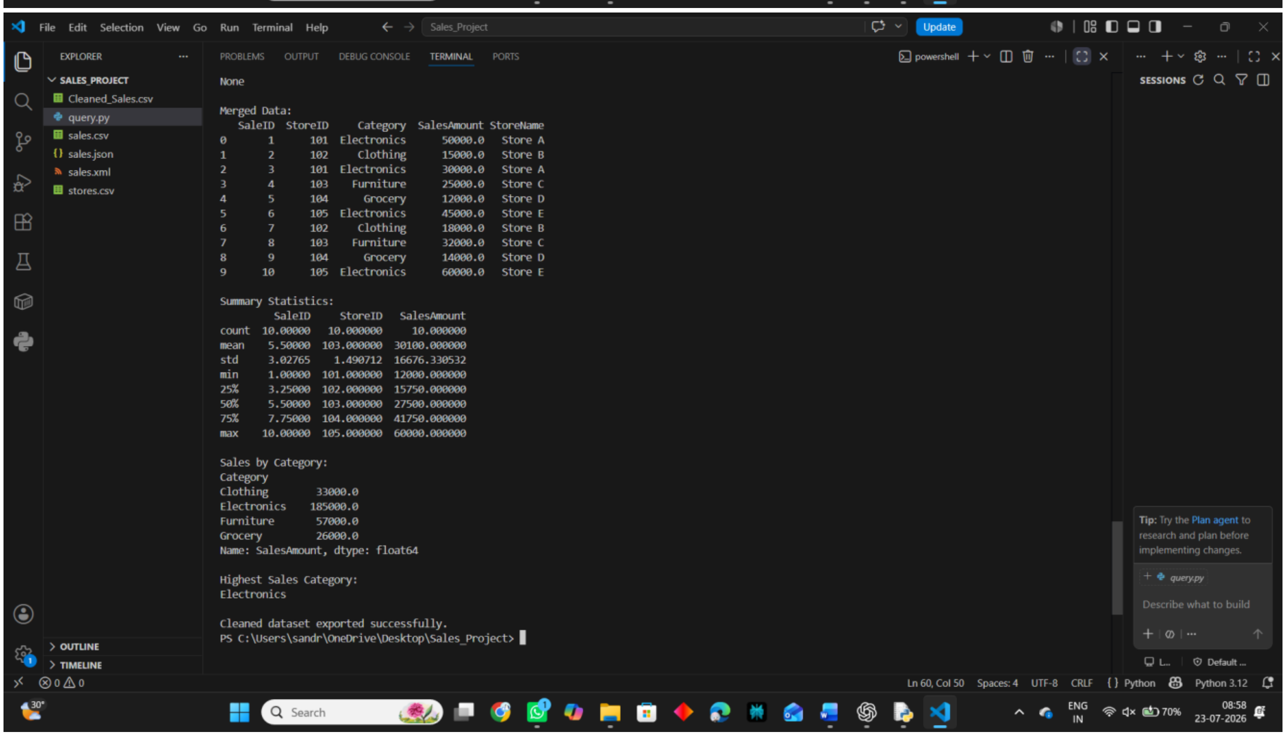
SaleID	StoreID	Category	SalesAmount t
1	101	Electronics	50000
2	102	Clothing	15000
3	101	Electronics	30000

4	103	Furniture	25000

Stores.csv

StoreID	StoreName
101	Store A
102	Store B
103	Store c





54. Hospital Patient Records

Files Required

- Patients.csv
- Doctors.csv

- Patients.json
- Patients.xml

```
import pandas as pd

# Read CSV files
patients = pd.read_csv("patients.csv")
doctors = pd.read_csv("doctors.csv")

# Read JSON file
patients_json = pd.read_json("patients.json")

# Read XML file
patients_xml = pd.read_xml("patients.xml", parser="etree")

# Inspect dataset
print("Patients Data:")
print(patients.head())

print("\nDataset Information:")
print(patients.info())

# Handle missing values
patients.fillna("Unknown", inplace=True)

# Remove duplicate records
patients.drop_duplicates(inplace=True)

# Validate data types
patients["Age"] = patients["Age"].astype(int)

# Merge Patients and Doctors using DoctorID
merged = pd.merge(patients, doctors, on="DoctorID")

print("\nMerged Data:")
print(merged)

# Summary statistics
print("\nSummary Statistics:")
print(merged.describe())

# Aggregation
summary = merged.groupby("Department")["Age"].mean()

print("\nAverage Age by Department:")
```

```python
    print(summary)

    # Department with highest average age
    print("\nDepartment with Highest Average Age:")
    print(summary.idxmax())

    # Export cleaned dataset
    merged.to_csv("Cleaned_Patients.csv", index=False)

    print("\nCleaned dataset exported successfully.")
```

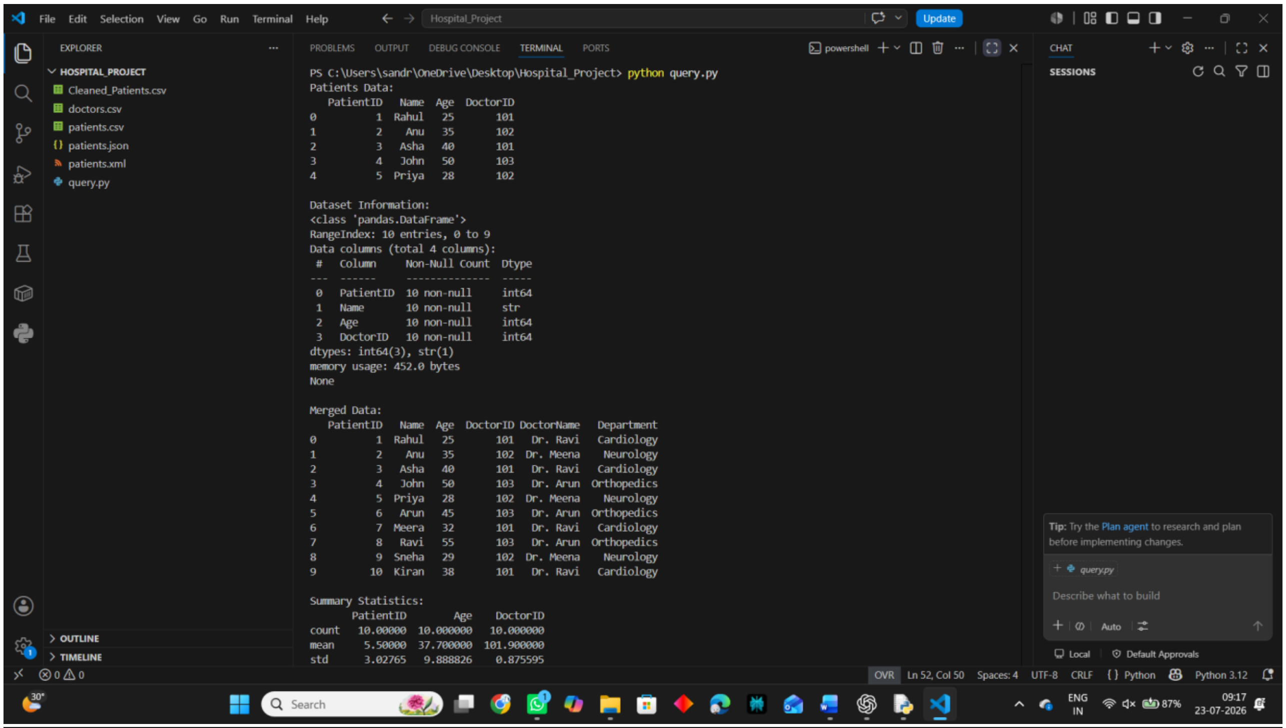
Sample Data

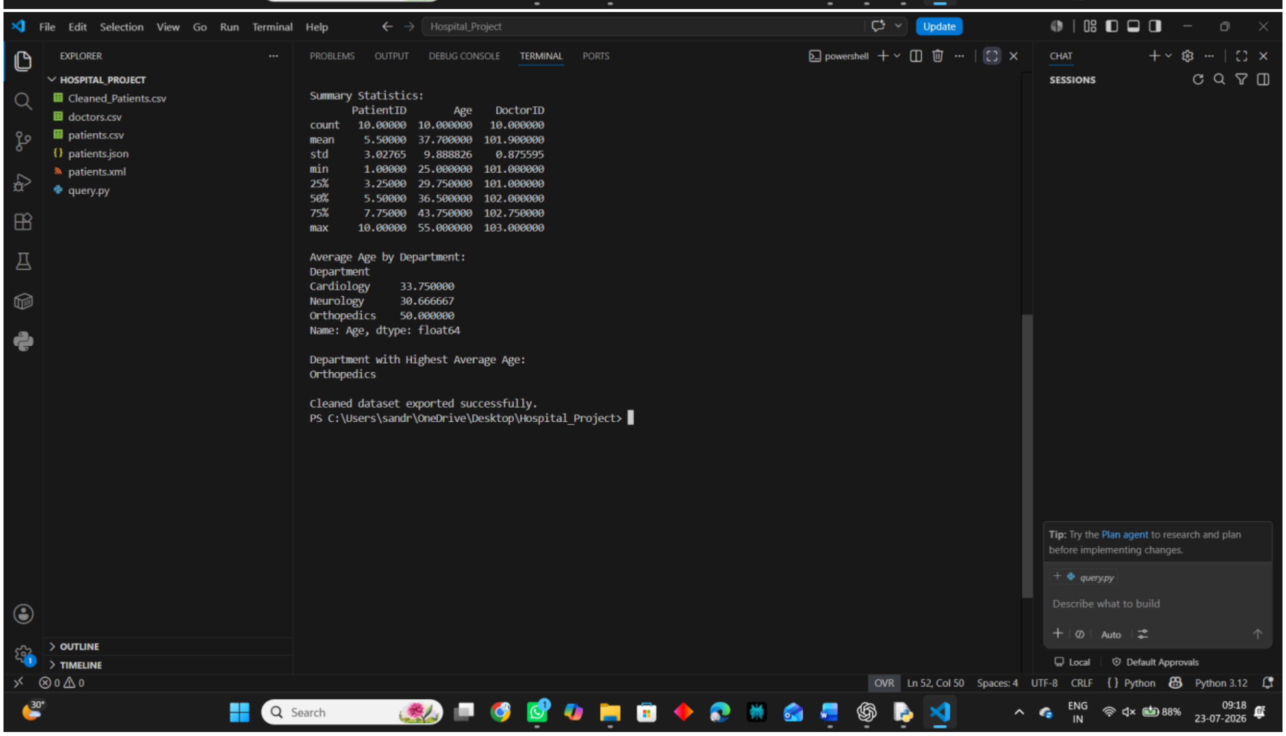
Patients.csv

PatientID	Name	Age	DoctorID
1	Rahul	25	101
2	Anu	35	102
3	Asha	40	101
4	John	50	103

Doctors.csv

DoctorID	DoctorName	Department
101	Dr. Ravi	Cardiology
102	Dr. Meena	Neurology
103	Dr. Arun	Orthopedics


PS C:\Users\sandr\OneDrive\Desktop\Hospital_Project> python query.py
Patients Data:
PatientID Name Age DoctorID
0 1 Rahul 25 101
1 2 Anu 35 102
2 3 Asha 40 101
3 4 John 50 103
4 5 Priya 28 102
Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
Column Non-Null Count Dtype
0 PatientID 10 non-null int64
1 Name 10 non-null str
2 Age 10 non-null int64
3 DoctorID 10 non-null int64
dtypes: int64(3), str(1)
memory usage: 452.0 bytes
None
Merged Data:
PatientID Name Age DoctorID DoctorName Department
0 1 Rahul 25 101 Dr. Ravi Cardiology
1 2 Anu 35 102 Dr. Meena Neurology
2 3 Asha 40 101 Dr. Ravi Cardiology
3 4 John 50 103 Dr. Arun Orthopedics
4 5 Priya 28 102 Dr. Meena Neurology
5 6 Arun 45 103 Dr. Arun Orthopedics
6 7 Meera 32 101 Dr. Ravi Cardiology
7 8 Ravi 55 103 Dr. Arun Orthopedics
8 9 Sneha 29 102 Dr. Meena Neurology
9 10 Kiran 38 101 Dr. Ravi Cardiology
Summary Statistics:
PatientID Age DoctorID
count 10.00000 10.000000 10.000000
mean 5.50000 37.700000 101.900000
std 3.02765 9.888826 0.875595


Summary Statistics:
PatientID Age DoctorID
count 10.00000 10.000000 10.000000
mean 5.50000 37.700000 101.900000
std 3.02765 9.888826 0.875595
min 1.00000 25.000000 101.000000
25% 3.25000 29.750000 101.000000
50% 5.50000 36.500000 102.000000
75% 7.75000 43.750000 102.750000
max 10.00000 55.000000 103.000000
Average Age by Department:
Department
Cardiology 33.750000
Neurology 30.666667
Orthopedics 50.000000
Name: Age, dtype: float64
Department with Highest Average Age:
Orthopedics
Cleaned dataset exported successfully.
PS C:\Users\sandr\OneDrive\Desktop\Hospital_Project>